

Thomson Scattering

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These notes use the [conventions](#) fixed earlier in the course, and build directly on the covariant radiation field derived in [Radiation from a Moving Charge](#) and the energy flux introduced in [Energy of the Electromagnetic Field](#). The goal is to go from the covariant field of an accelerated charge, through its non-relativistic (local rest-frame) limit, to the classical differential and total cross sections for scattering of light by a free charge.

1 Setup

A free charge q of mass m (an electron, in practice) sits at rest at the origin and is illuminated by a monochromatic plane wave arriving from a distant source along the fixed direction $\hat{\mathbf{k}}$,

$$\mathbf{E}_{\text{inc}}(t) = E_0 \hat{\mathbf{e}} \cos(\omega t), \quad \mathbf{B}_{\text{inc}}(t) = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}_{\text{inc}}(t),$$

with $\hat{\mathbf{e}} \perp \hat{\mathbf{k}}$, evaluated at the charge's location. This is exactly the radiation-field structure found in [Radiation from a Moving Charge](#), $\mathbf{B}_{\text{rad}} = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}_{\text{rad}}$ with $\hat{\mathbf{n}} \cdot \mathbf{E}_{\text{rad}} = 0$: a plane wave is simply what any radiation field looks like once its distant source is out of the picture, with $\hat{\mathbf{k}}$ now playing the role of the *incoming* propagation direction, in place of the outgoing $\hat{\mathbf{n}}$ used there to point from source to observer, and $\hat{\mathbf{e}} \perp \hat{\mathbf{k}}$ the same transversality condition as $\hat{\mathbf{n}} \cdot \mathbf{E}_{\text{rad}} = 0$. In particular $|\mathbf{B}_{\text{inc}}| = |\mathbf{E}_{\text{inc}}|/c$ follows immediately, [exactly as for any radiation field](#). Two approximations define the classical (Thomson) regime:

- **non-relativistic motion:** the charge's response velocity $v \ll c$, so terms of order v/c are dropped;
- **no radiation reaction:** the energy radiated per cycle is a small fraction of the charge's kinetic energy, so the incident field alone drives the motion.

2 Equation of motion

The Lorentz force on the charge is $q(\mathbf{E}_{\text{inc}} + \mathbf{v} \times \mathbf{B}_{\text{inc}})$. Since $|\mathbf{B}_{\text{inc}}| = |\mathbf{E}_{\text{inc}}|/c$, the magnetic term is smaller than the electric term by a factor $v/c \ll 1$ and is dropped, leaving

$$m\dot{\mathbf{v}} = q\mathbf{E}_{\text{inc}}(t), \quad \dot{\mathbf{v}}(t) = \frac{q}{m} E_0 \hat{\mathbf{e}} \cos(\omega t).$$

The charge oscillates along the fixed direction $\hat{\mathbf{e}}$ with acceleration amplitude $a_0 \equiv qE_0/m$; since $a_0/\omega \ll c$ in the non-relativistic regime, the charge's excursion is tiny compared to the wavelength, and it radiates essentially from a fixed point (the dipole approximation).

3 From the covariant field to the local frame

The [radiation field](#) of an accelerated charge, valid for any velocity, was written covariantly as

$$F_{\mu\nu}^{\text{rad}} = \frac{\mu_0 q}{4\pi(u \cdot \Delta x)^2} \left[\Delta x_\mu a_\nu - \Delta x_\nu a_\mu - \frac{\Delta x \cdot a}{u \cdot \Delta x} (\Delta x_\mu u_\nu - \Delta x_\nu u_\mu) \right]_{\tau=\tau_r},$$

with $a^\mu = du^\mu/d\tau$. Here this is specialized to a charge that is instantaneously at rest, as is the case for the non-relativistic oscillation driven by the incident wave.

At the instant the charge is at rest, $u^\mu = (1, \mathbf{0})$ and, since $u_\mu a^\mu = 0$ forces $a^0 = 0$ there, $a^\mu = (0, \dot{\mathbf{v}}/c)$ exactly (this is the same identification used for the [Larmor formula](#)). With $\Delta x^\mu = (R, R\hat{\mathbf{n}})$, $R = |\mathbf{x}|$,

$$u \cdot \Delta x = -R, \quad \Delta x \cdot a = R \frac{\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}}{c}.$$

Substituting into $F_{0i}|_{\text{rad}} = -E_i/c$ and simplifying (the two $1/R$ -carrying terms combine; the $\Delta x_\mu u_\nu$ term contributes $R\hat{n}_i$ using $u_i = 0, u_0 = -1$) gives

$$\mathbf{E}_{\text{rad}}(\mathbf{x}, t) = \frac{\mu_0 q}{4\pi R} \left[\dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}} \right]_{t_r} = -\frac{\mu_0 q}{4\pi R} \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}}) \Big|_{t_r},$$

the familiar non-relativistic dipole radiation field, transverse to $\hat{\mathbf{n}}$ by construction. Writing χ for the angle between $\hat{\mathbf{n}}$ and $\dot{\mathbf{v}}$, its magnitude is

$$|\mathbf{E}_{\text{rad}}| = \frac{\mu_0 q |\dot{\mathbf{v}}|}{4\pi R} \sin \chi.$$

4 Radiated power and a consistency check

In the radiation zone, $u = \varepsilon_0 E^2$ and $\mathbf{S} = cu \hat{\mathbf{n}}$ (from the [Radiation zone](#) section), so the power radiated per solid angle is

$$\frac{dP}{d\Omega} = R^2 \mathbf{S} \cdot \hat{\mathbf{n}} = R^2 c \varepsilon_0 |\mathbf{E}_{\text{rad}}|^2 = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c} \sin^2 \chi,$$

using $1/(4\pi\varepsilon_0) = \mu_0 c^2/(4\pi)$. As a check, integrating over solid angle with $\int \sin^2 \chi d\Omega = 8\pi/3$ recovers exactly the non-relativistic Larmor formula already derived,

$$P = \int \frac{dP}{d\Omega} d\Omega = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c} \cdot \frac{8\pi}{3} = \frac{\mu_0 q^2 \dot{v}^2}{6\pi c},$$

matching [the boxed result](#) $P = \mu_0 q^2 \dot{v}^2/(6\pi c)$ obtained independently from the covariant radiation tensor.

5 Differential cross section

The incident wave carries energy density $u_{\text{inc}} = \varepsilon_0 E_{\text{inc}}^2$ and flux $S_{\text{inc}} = cu_{\text{inc}}$ (same radiation-zone relations, since a plane wave is locally the same kind of transverse field); its time average over a cycle is

$$\langle S_{\text{inc}} \rangle = c\varepsilon_0 \langle E_{\text{inc}}^2 \rangle = \frac{1}{2} c\varepsilon_0 E_0^2.$$

What a cross section measures. A distant detector at angle (θ, ϕ) subtending solid angle $d\Omega$ records scattered power $\langle dP/d\Omega \rangle d\Omega$, independent of its distance (since $dP/d\Omega$ was built from the R -independent radiation field found above). The incident beam is instead characterized by its flux $\langle S_{\text{inc}} \rangle$, a power per unit *area* crossing a surface perpendicular to $\hat{\mathbf{k}}$. Dividing the two cancels the (arbitrary) incident intensity and turns “power per solid angle” into “area per solid angle” — hence the name *cross section*:

$$\frac{d\sigma}{d\Omega} \equiv \frac{\langle dP/d\Omega \rangle}{\langle S_{\text{inc}} \rangle}, \quad [d\sigma/d\Omega] = \text{area}.$$

Concretely, $d\sigma$ is the cross-sectional area of the incident beam that carries exactly the power the charge re-radiates into $d\Omega$; integrated over all directions ([below](#)), the total cross section σ is the area of a flat, perfectly absorbing target that would intercept, head-on, the same total power the charge actually scatters in every direction. It is the effective size the charge presents to the beam. Because both $\langle dP/d\Omega \rangle$ and $\langle S_{\text{inc}} \rangle$ scale as E_0^2 , this ratio is independent of the incident intensity: it depends only on the scattering geometry and on q and m , which is what makes a cross section a property of the charge (or process) rather than of any particular beam. Using $\dot{\mathbf{v}} = (q/m)\mathbf{E}_{\text{inc}}$ from the equation of motion, $\langle \dot{v}^2 \rangle = (q/m)^2 \langle E_{\text{inc}}^2 \rangle = (q/m)^2 E_0^2/2$, so

$$\left\langle \frac{dP}{d\Omega} \right\rangle = \frac{\mu_0 q^2}{16\pi^2 c} \cdot \frac{q^2 E_0^2}{2m^2} \sin^2 \chi,$$

and dividing by $\langle S_{\text{inc}} \rangle = c\varepsilon_0 E_0^2/2$, with $1/\varepsilon_0 = \mu_0 c^2$,

$$\frac{d\sigma}{d\Omega} = \left(\frac{\mu_0 q^2}{4\pi m} \right)^2 \sin^2 \chi.$$

Defining the **classical radius** of the charge,

$$r_q \equiv \frac{\mu_0 q^2}{4\pi m} = \frac{q^2}{4\pi\varepsilon_0 m c^2},$$

this is

$$\boxed{\frac{d\sigma}{d\Omega} = r_q^2 \sin^2 \chi},$$

with χ the angle between the observation direction $\hat{\mathbf{n}}$ and the (fixed) oscillation direction $\hat{\mathbf{e}}$: scattering is strongest broadside to the oscillation and vanishes along it, exactly like a driven dipole antenna.

6 Unpolarized incident light

For unpolarized incident light, average $\sin^2 \chi$ over the two independent transverse polarizations. Take $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ and, without loss of generality, the observation direction in the x - z plane at scattering angle θ from $\hat{\mathbf{k}}$, $\hat{\mathbf{n}} = \sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{z}}$. For $\hat{\mathbf{e}} = \hat{\mathbf{x}}$, $\sin^2 \chi = 1 - (\hat{\mathbf{n}} \cdot \hat{\mathbf{x}})^2 = \cos^2 \theta$; for $\hat{\mathbf{e}} = \hat{\mathbf{y}}$, $\hat{\mathbf{n}} \cdot \hat{\mathbf{y}} = 0$ so $\sin^2 \chi = 1$. Averaging the two,

$$\langle \sin^2 \chi \rangle = \frac{\cos^2 \theta + 1}{2},$$

so that

$$\boxed{\left. \frac{d\sigma}{d\Omega} \right|_{\text{unpol}} = \frac{r_q^2}{2} (1 + \cos^2 \theta)},$$

the classical Thomson formula, with θ now the angle between the scattered and incident directions.

7 Total cross section

Integrating over solid angle, with $x = \cos \theta$,

$$\sigma_T = \frac{r_q^2}{2} \int_0^{2\pi} d\phi \int_{-1}^1 (1 + x^2) dx = \pi r_q^2 \left[x + \frac{x^3}{3} \right]_{-1}^1 = \pi r_q^2 \cdot \frac{8}{3},$$

$$\boxed{\sigma_T = \frac{8\pi}{3} r_q^2}.$$

For an electron ($q = e$, $m = m_e$), r_q is the classical electron radius and $\sigma_T \approx 6.65 \times 10^{-29} \text{ m}^2$ is the Thomson cross section.

8 Validity

This is a purely classical, elastic calculation: the scattered light has the same frequency ω as the incident light, with no recoil. It holds provided:

- $v \ll c$ throughout the motion (already used to drop the magnetic force and to evaluate a^μ at $u^\mu = (1, \mathbf{0})$);
- the photon energy $\hbar\omega$ is much smaller than mc^2 , so that the quantum recoil corrections of Compton scattering are negligible;
- the charge is genuinely free (or driven far from any resonance), so that no restoring force competes with the incident field in the equation of motion.

9 Exercises

9.1 Exercise 1: Dropping the Magnetic Force

a) Using $|\mathbf{B}_{\text{inc}}| = |\mathbf{E}_{\text{inc}}|/c$, estimate the ratio $|q\mathbf{v} \times \mathbf{B}_{\text{inc}}|/|q\mathbf{E}_{\text{inc}}|$ in terms of v/c , confirming that it is suppressed for non-relativistic motion.

b) Using $v \sim a_0/\omega$ for the oscillation velocity scale, with $a_0 = qE_0/m$, express v/c in terms of q, E_0, m, ω, c , and check that it is indeed small for visible light ($\omega \sim 10^{15} \text{ s}^{-1}$) incident on a free electron at ordinary laboratory intensities.

9.2 Exercise 2: The Local-Frame Radiation Field (Essential)

a) Starting from $F_{\mu\nu}^{\text{rad}}$ specialized to $u^\mu = (1, \mathbf{0})$, with $\Delta x^\mu = (R, R\hat{\mathbf{n}})$ and $\Delta x \cdot a = R\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}/c$, write out F_{0i}^{rad} term by term and show that the two $1/R$ -carrying contributions combine into

$$\mathbf{E}_{\text{rad}} = \frac{\mu_0 q}{4\pi R} [\dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}}].$$

b) Verify the equivalent form $-\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}}) = \dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}}$ using the BAC-CAB rule.

9.3 Exercise 3: Larmor Consistency Check (Essential)

a) Verify $\int \sin^2 \chi \, d\Omega = 8\pi/3$ by direct integration in spherical coordinates, with χ the polar angle measured from $\dot{\mathbf{v}}$.

b) Substitute this into $dP/d\Omega$ and confirm that the result reproduces exactly the Larmor formula $P = \mu_0 q^2 \dot{v}^2 / (6\pi c)$ derived covariantly in [Radiation from a Moving Charge](#), matching the numerical prefactor as well as the power of $\dot{v}$.

9.4 Exercise 4: The Differential Cross Section, Step by Step (Essential)

a) Using $\dot{\mathbf{v}} = (q/m)\mathbf{E}_{\text{inc}}$ and $\langle E_{\text{inc}}^2 \rangle = E_0^2/2$, verify $\langle \dot{v}^2 \rangle = (q/m)^2 E_0^2/2$ and substitute it into $\langle dP/d\Omega \rangle$ to reproduce the boxed intermediate expression given in the notes.

b) Divide by $\langle S_{\text{inc}} \rangle$ and, using $1/\varepsilon_0 = \mu_0 c^2$, verify explicitly that every factor of c , μ_0 , and m combines to leave $d\sigma/d\Omega = r_q^2 \sin^2 \chi$, an area.

9.5 Exercise 5: Unpolarized Averaging in a General Direction

a) Starting from $\cos \chi = \hat{\mathbf{n}} \cdot \hat{\boldsymbol{\epsilon}}$, verify $\sin^2 \chi = 1 - (\hat{\mathbf{n}} \cdot \hat{\boldsymbol{\epsilon}})^2$.

b) Using the rotational symmetry of the setup about $\hat{\mathbf{k}}$, argue that the unpolarized $d\sigma/d\Omega$ can only depend on the polar angle θ between $\hat{\mathbf{n}}$ and $\hat{\mathbf{k}}$, never on the azimuthal angle of $\hat{\mathbf{n}}$ — so it suffices to check the formula for $\hat{\mathbf{n}}$ in a single plane containing $\hat{\mathbf{k}}$, as the notes do.

9.6 Exercise 6: The Total Cross Section (Essential)

a) Verify $\int_0^{2\pi} d\phi \int_{-1}^1 (1+x^2) dx = 8\pi/3$ step by step, and hence confirm $\sigma_T = \frac{8\pi}{3} r_q^2$.

b) Using $r_q = q^2/(4\pi\varepsilon_0 mc^2)$ and $e = 1.6 \times 10^{-19} \text{ C}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, verify numerically that $\sigma_T \approx 6.65 \times 10^{-29} \text{ m}^2$ for an electron.

Feel free to create issues, ask questions, or suggest improvements in the [GitHub repository](#).